

# Resilient and sustainable supplier selection: a chance-constrained approach under disruption risk

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Laila Messaoudi

University of Gabes, Gabes, Tunisia

Faiza Hamdi

Supply Chain Management Department, University of Jeddah,  
Jeddah, Saudi Arabia, and

Jalel Euchì

Department of Quantitative Methods, University of Sousse, Sousse, Tunisia and  
OLID Laboratory, Higher Institute of Industrial Management, University of Sfax,  
Sfax, Tunisia

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## Abstract

**Purpose** – In today's volatile business environment, firms are required to balance cost efficiency, sustainability and operational resilience to sustain competitiveness. Supplier selection plays a pivotal role in this context; however, traditional models often fail to capture disruption risks caused by geopolitical tensions, health crises or natural disasters. This study aims to address this gap by developing a framework that explicitly incorporates disruption probabilities into sustainable supplier selection.

**Design/methodology/approach** – A chance-constrained goal programming (GP) model, enriched with Value-at-Risk (VaR) metrics, is proposed. The framework integrates disruption probabilities, demand variability, procurement costs and sustainability performance indicators to jointly optimize supplier choice. A real-world case from the automotive sector is used to demonstrate the applicability and robustness of the model under varying risk tolerance levels.

**Findings** – The empirical results reveal that the model consistently selects resilient supplier portfolios even under high uncertainty. It ensures cost control, safeguards sustainability thresholds and prevents excessive dependence on a small number of suppliers. The analysis highlights that incorporating probabilistic disruption modeling significantly improves procurement stability compared to conventional approaches.

**Originality/value** – Unlike prior studies that consider sustainability and disruption risk separately, this research offers an integrated framework that simultaneously accounts for financial, environmental and disruption-related concerns. By embedding disruption probabilities and diversification constraints into supplier selection, the study addresses a critical gap in procurement planning and contributes a novel methodological advancement for sustainable sourcing under uncertainty.

**Keywords** Sustainability, Disruption risk, Supplier selection, Chance-constrained goal programming

**Paper type** Research article

## 1. Introduction

Managing supply chains (SC) has become a critical business priority due to their direct influence on economic performance, environmental responsibility, and organizational resilience (Ivanov, 2022a, b). In today's global marketplace, firms are increasingly exposed to disruptions arising from natural disasters, health crises, geopolitical tensions, and logistical bottlenecks, all of which can trigger severe delays, cost overruns, and reputational damage.

In an era characterized by growing economic volatility, stricter environmental regulations, and recurrent supply chain disruptions, companies are increasingly compelled to reassess their

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supplier evaluation frameworks (Chai and Ngai, 2020; Ivanov, 2022a, b). Supplier-related decisions must now simultaneously account for cost efficiency, sustainability performance, and exposure to disruption risk, as overlooking any of these dimensions can compromise both competitiveness and operational continuity (Kusi-Sarpong *et al.*, 2019; Karakoç *et al.*, 2023). These priorities align with global trends that emphasize the integration of resilience and sustainability into procurement practices, particularly in industries where supply interruptions can have cascading economic and reputational consequences (Gökler and Boran, 2025; Rashidi *et al.*, 2020). This study therefore, emphasizes three interrelated dimensions of supplier selection: minimizing procurement costs, promoting sustainability, and reducing exposure to disruption risk, which together reflect the strategic necessity of remaining competitive in an uncertain and regulation-driven business environment.

Despite their importance, these dimensions are often addressed in isolation within existing models. Many prior studies emphasize either cost-efficiency or sustainability, while the treatment of disruption risk particularly under uncertainty remains underdeveloped. This lack of integration creates a methodological gap, especially in industries like automotive manufacturing, where supplier reliability and environmental performance are both strategic imperatives. To address this challenge, the present research proposes a decision-making framework that integrates these dimensions using a chance-constrained GP model, supported by Value-at-Risk (VaR) metrics. This allows for a more robust and risk-aware supplier portfolio optimization under uncertainty.

This transition is further reinforced by global policy directives that encourage firms to account for environmental and social impacts throughout their supply chains. Beyond regulatory compliance, these directives create strategic incentives for companies to embed sustainability and resilience into sourcing practices. As a result, supplier selection is no longer limited to cost-based criteria but increasingly viewed as a multi-dimensional decision that balances financial efficiency, environmental performance, and the capacity to withstand unexpected disruptions.

In light of these challenges, a growing number of studies have explored how to embed sustainability into supplier evaluation. Recent reviews, such as Rashidi *et al.* (2020), have classified sustainable supplier selection (SSS) research into economic, environmental, and social dimensions, highlighting the strategic role of supplier choice in sustainable procurement. Others, like Gökler and Boran (2025), have emphasized the importance of selecting suppliers who can withstand disruption risks stemming from both natural and human causes, especially in critical industries like automotive manufacturing.

Despite these valuable contributions, many of these approaches either prioritize sustainability or address risk in isolation. Only a limited number of studies attempt to handle both simultaneously, and fewer still incorporate probabilistic models of disruption risk into the decision-making framework. This underscores the need for a unified, optimization-based model that captures both sustainability criteria and supply chain uncertainty in supplier selection and order allocation.

There are situations where decisions are clear-cut, but there are also more complex ones where there may be conflicting factors that need to be considered simultaneously with different priorities and degrees of relevance in order to make a rational decision (Hamdi *et al.*, 2023; Euchi and Yassine, 2023; Sahoo Goswami, 2024; Messaoudi *et al.*, 2017; Messaoudi, 2024).

While various techniques such as fuzzy Multi-Criteria Decision-Making (MCDM) models, classical Goal Programming (GP), and Data Envelopment Analysis (DEA)-based methods have contributed to sustainable supplier selection, many of them fall short in two key areas: first, they often optimize supplier choice without considering order allocation in detail; second, they typically lack a unified framework for integrating sustainability with stochastic risk modeling. For example, approaches such as fuzzy VIKOR (VISeKriterijumska Optimizacija I Kompromisno Resenje, meaning Multi-Criteria Optimization and Compromise Solution) or fuzzy TOPSIS (Technique for Order Preference by Similarity to the Ideal Solution) deal with uncertainty in a qualitative manner, but they do not incorporate

probabilistic risk thresholds such as those commonly applied in financial decision-making (Aouadni and Euchi, 2022; Euchi *et al.*, 2019).

Based on these questions, our main contributions are summarized as follows. Moreover, few studies have attempted to quantify disruption-related uncertainty using metrics such as Value-at-Risk (VaR), which is critical for procurement decisions under high volatility.

These gaps are especially critical in sectors like automotive manufacturing, where procurement decisions involve balancing stringent sustainability goals with supply continuity, often under uncertainty. The current study responds to this challenge by developing a quantitative, risk-aware framework that integrates supplier selection with order allocation, using chance-constrained optimization and VaR modeling. By doing so, we bridge methodological and practical gaps in sustainable supplier selection under disruption risk.

To guide this investigation, the study addresses the following scientific questions:

- (1) *How can firms simultaneously address sustainability performance, cost efficiency, and disruption risk in supplier selection under uncertainty?*
- (2) *What is the impact of using probabilistic constraints and Value-at-Risk (VaR) modeling in shaping robust and sustainable supplier portfolios?*
- (3) *How does a chance-constrained GP approach perform in a real-world setting when varying risk tolerance levels are introduced?*

Based on these questions, the study makes several contributions. First, it develops a multi-objective chance-constrained goal programming (GP) model that simultaneously integrates procurement cost, sustainability performance, and disruption risk through Value-at-Risk (VaR)-based constraints. Second, the proposed framework extends existing approaches by enabling risk-aware order allocation, thereby overcoming the limitation of many methods that focus only on supplier rankings without addressing quantity distribution. Third, the model is validated through a real-world case study from the automotive sector, and a sensitivity analysis is conducted to evaluate its responsiveness under different risk tolerance scenarios. To the best of our knowledge, this work represents the first application of a chance-constrained formulation incorporating VaR within the context of sustainable supplier selection and order allocation. Finally, by bridging the gap between supply chain resilience, environmental performance, and quantitative risk modeling, the research provides both theoretical and managerial insights into sustainable procurement under uncertainty.

By closing the methodological gap between supply chain resilience, environmental performance, and quantitative risk modeling, this research contributes to both the theory and practice of sustainable procurement under uncertainty.

The structure of this paper is as follows: Section 2 presents a thorough literature review on sustainable supplier selection and supply chain risk management. Section 3 outlines the research methodology and details the mathematical model formulation. Section 4 discusses the case study and results, while Section 5 concludes by summarizing key insights and proposing future research directions.

## 2. Literature review

The literature on supplier-related decision-making has evolved considerably in recent years, especially with the growing emphasis on sustainable procurement, risk mitigation, and dynamic resource allocation. This review covers three key strands: (1) sustainable supplier selection (SSS), (2) order allocation in uncertain environments, and (3) hybrid and risk-aware decision-making frameworks.

### 2.1 Sustainable supplier selection (SSS)

Driven by regulatory, environmental, and ethical imperatives, SSS has become a cornerstone of procurement strategy across industries. At its core, it integrates environmental, social, and

economic factors into supplier evaluation, often operationalized through multi-criteria decision-making (MCDM) tools.

Researchers such as [Dweiri et al. \(2016\)](#), [Costa et al. \(2018\)](#), and [Karakoç et al. \(2023\)](#) have utilized frameworks like ELECTRE, TOPSIS, and hybrid BWM-TOPSIS models to rank suppliers across sustainability criteria. [Koc et al. \(2023\)](#) further extend this view by adding lean management, knowledge integration, and innovation as decision attributes.

A wide range of approaches has been developed to support sustainable supplier selection. Early contributions, such as [Dweiri et al. \(2016\)](#), [Costa et al. \(2018\)](#), and [Liu et al. \(2019\)](#), applied multi-criteria decision-making (MCDM) techniques like ELECTRE and TOPSIS to capture trade-offs among quality, delivery, and environmental performance. More recent studies have emphasized hybrid formulations, including [Karakoç et al. \(2023\)](#), who combined BWM and [Koc et al. \(2023\)](#), who incorporated lean management and innovation into supplier evaluation. Other notable works have explored fuzzy and intuitionistic extensions to handle ambiguity, such as [Gupta et al. \(2019\)](#), [Stević et al. \(2020\)](#), and [Tohidi et al. \(2024\)](#). These studies collectively underline the growing importance of integrating environmental, social, and resilience-related dimensions into supplier evaluation frameworks, moving beyond traditional cost and quality criteria.

[Table 1](#) summarizes recent approaches and highlights their methodological diversity, ranging from classical GP to fuzzy logic and intuitionistic MCDM methods. Most models focus on supplier evaluation based on Triple Bottom Line (TBL) metrics. However, they often lack the capabilities to address dynamic conditions such as supply disruptions, capacity variability, or risk sensitivity.

Furthermore, recent reviews (e.g. [Karakoç et al., 2023](#); [Gidiagba et al., 2023](#)) underscore the growing popularity of hybrid approaches and fuzzy extensions, which better accommodate ambiguity in sustainability criteria. Still, these approaches typically assume a stable environment and often stop at supplier ranking, without proceeding to actual procurement or order allocation under real-world constraints.

## 2.2 Order allocation models under supply chain constraints

Order allocation, while closely linked to supplier selection, involves distinct optimization challenges. This body of work typically deals with assigning quantities to selected suppliers while respecting constraints such as capacity, discounts, delivery lead times, or service levels.

Order allocation has also been studied extensively, particularly in contexts where multiple suppliers must be coordinated under capacity and risk constraints. Seminal work by [Amid et al. \(2011\)](#) introduced fuzzy multi-objective programming to balance cost and service trade-offs. Subsequent contributions, such as [Cheraghalipour and Farsad \(2018\)](#) and [Bektur \(2020\)](#) examined allocation decisions under disruption risks, demand variability, and quantity discounts. More recent studies, including [Ahmadi and Amin \(2019\)](#) and [Alavi et al. \(2021\)](#), have integrated stochastic or dynamic elements into allocation models, reflecting real-world complexities such as fluctuating demand and sustainability requirements. Together, these works show that while order allocation is operationally critical, many models still treat risk deterministically and rarely integrate sustainability thresholds into the allocation process.

While such models are operationally relevant, they often treat risk deterministically and lack mechanisms to represent the stochastic nature of disruption events or performance fluctuations. Moreover, many do not provide a unified framework to optimize both supplier evaluation and order allocation simultaneously under risk.

## 2.3 Integrated frameworks incorporating risk and sustainability

A growing body of research has aimed to integrate sustainability, cost, and risk considerations into unified frameworks. For example, [Moheb-Alizadeh and Handfield \(2018, 2019\)](#) proposed stochastic models combining disruption scenarios with sustainability objectives, while [Jia et al. \(2020\)](#) developed a goal programming framework with chance constraints to address uncertainty in supplier performance. Other recent contributions, such as [Kellner and Utz \(2019\)](#) and

**Table 1.** Overview of key criteria and methods in sustainable supplier selection models

| Reference                         | Economic |   |   | Social |   | Environmental |                   | Resilience/<br>Innovation |    |    | Geographical<br>Distance | Methodology                         |
|-----------------------------------|----------|---|---|--------|---|---------------|-------------------|---------------------------|----|----|--------------------------|-------------------------------------|
|                                   | Q        | P | D | R      | F | PC            | Green<br>products | I                         | RM | LP |                          |                                     |
| Dweiri <i>et al.</i> (2016)       |          | ✓ | ✓ |        |   | ✓             |                   |                           |    |    | ✓                        | ELECTRE TRI-NC                      |
| Costa <i>et al.</i> (2018)        | ✓        | ✓ | ✓ |        | ✓ | ✓             |                   |                           | ✓  |    |                          |                                     |
| Liu <i>et al.</i> (2019)          | ✓        |   |   |        |   | ✓             |                   |                           |    |    |                          |                                     |
| Taherdoost <i>et al.</i> (2019)   | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    |    | ✓                        | MCDM                                |
| Yu <i>et al.</i> (2019)           | ✓        | ✓ | ✓ | ✓      | ✓ | ✓             |                   |                           |    |    |                          |                                     |
| Ahmadi <i>et al.</i> (2019)       |          | ✓ | ✓ |        | ✓ | ✓             |                   |                           |    |    |                          | Sustainability Innovation Framework |
| Kusi-Sarpong <i>et al.</i> (2019) | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    | ✓  |                          |                                     |
| Kannan <i>et al.</i> (2020)       | ✓        | ✓ | ✓ |        |   |               | ✓                 |                           |    |    |                          |                                     |
| Stević <i>et al.</i> (2020)       | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    |    | ✓                        | Goal programming                    |
| Jia <i>et al.</i> (2020)          | ✓        | ✓ |   |        |   |               |                   |                           |    |    |                          |                                     |
| Alavi <i>et al.</i> (2021)        | ✓        | ✓ | ✓ |        | ✓ |               | ✓                 |                           |    |    |                          |                                     |
| Afrasiabi <i>et al.</i> (2022)    | ✓        | ✓ | ✓ |        |   | ✓             | ✓                 |                           |    |    |                          | Fuzzy TOPSIS                        |
| Baki <i>et al.</i> (2022)         |          | ✓ | ✓ | ✓      |   |               |                   |                           |    |    |                          |                                     |
| Koc <i>et al.</i> (2023)          | ✓        | ✓ |   | ✓      |   | ✓             | ✓                 | ✓                         |    | ✓  |                          |                                     |
| Karakoç <i>et al.</i> (2023)      | ✓        | ✓ | ✓ | ✓      | ✓ | ✓             | ✓                 |                           | ✓  |    |                          | Extended TBL framework              |
| Ben Abdallah <i>et al.</i> (2024) | ✓        | ✓ | ✓ | ✓      | ✓ | ✓             | ✓                 |                           | ✓  |    |                          |                                     |
| Ghazvinian <i>et al.</i> (2024)   | ✓        | ✓ |   | ✓      |   | ✓             | ✓                 | ✓                         |    |    |                          |                                     |
| Gupta <i>et al.</i> (2019)        | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    | ✓  |                          | Hybrid TOPSIS-BWM                   |
| Tohidi <i>et al.</i> (2024)       | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    | ✓  |                          |                                     |
| Ajalli (2024)                     | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    | ✓  |                          |                                     |
| Varchandi <i>et al.</i> (2024)    | ✓        | ✓ | ✓ | ✓      |   | ✓             | ✓                 |                           |    | ✓  |                          | Fuzzy MCDM                          |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          |                                     |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          | Z-Numbers                           |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          |                                     |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          | Fuzzy Delphi-BWM-TOPSIS             |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          |                                     |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          | Best-Worst Method + Fuzzy TOPSIS    |
|                                   |          |   |   |        |   |               |                   |                           |    |    |                          |                                     |

**Note(s):** ✓ indicates the criterion is considered in the model. Abbreviations: Q = Quality, P = Price, D = Delivery, R = Reliability, F = Flexibility, PC = Pollution Control, I = Innovation, RM = Risk Management, LP = Lean Principles. TBL = Triple Bottom Line; IF-TOPSIS = Intuitionistic Fuzzy TOPSIS; LARGS = Large-Scale Group Decision-Making

**Source(s):** Created by authors based on the review

[Euchi \(2020\)](#), incorporated financial risk metrics like Value-at-Risk (VaR) into supplier portfolio optimization, thereby linking procurement with risk management practices. More recent work has also explored the role of emerging technologies; for instance, [Yilmaz et al. \(2025a, 2025b\)](#) highlighted the potential of machine learning to predict disruptions and dynamically adjust sourcing. Similarly, [Gökler and Boran \(2025\)](#) proposed a combined D-AHP and DEMATEL approach to account for both sustainability and resilience in supplier selection. Collectively, these studies demonstrate a shift from isolated models toward integrated, risk-aware frameworks that reflect the realities of modern supply chain management.

The use of financial risk tools such as Value-at-Risk (VaR), as introduced in the procurement context by [Kellner and Utz \(2019\)](#), represents a key methodological advancement. These models enable procurement planners to quantify potential loss exposure within defined confidence intervals, thereby aligning procurement with risk appetite.

Nevertheless, despite these contributions, few studies provide a multi-objective optimization structure that unifies probabilistic risk, sustainability thresholds, and cost minimization coherently. In particular, the portfolio-level treatment of suppliers wherein disruption risk and sustainability scores are jointly optimized is rarely addressed in existing works.

Beyond traditional optimization frameworks, recent studies have emphasized the role of technological innovation in enhancing supply chain resilience. For example, [Euchi \(2021\)](#) explored the feasibility of using drones for medical supply distribution during pandemic-related disruptions, highlighting how emerging technologies can complement supplier selection models by providing additional flexibility under uncertainty.

Finally, the rise of artificial intelligence and machine learning has begun influencing supplier decision-making. Studies such as [Yilmaz et al. \(2025a, b\)](#) highlight the potential of predictive analytics to anticipate supply disruptions and optimize sourcing dynamically. Yet, these approaches are still emerging and not fully integrated into robust mathematical optimization models.

While considerable progress has been made in SSS and order allocation, existing models often address these dimensions in isolation. Many MCDM-based approaches emphasize supplier scoring without considering quantity allocation, while operational models overlook sustainability and risk. Even integrated models tend to rely on deterministic assumptions or simplistic risk approximations.

This study aims to bridge these gaps by proposing a stochastic, chance-constrained GP model that unites sustainability metrics, cost parameters, and probabilistic risk modeling (via VaR). In doing so, it offers both theoretical novelty and practical relevance for sustainable procurement under uncertainty.

### 3. Research methodology

To develop a robust supplier selection framework, it is essential to consider both sustainability and risk comprehensively. Traditional approaches often overlook the dynamic nature of disruptions and the probabilistic uncertainties associated with supplier performance. By integrating advanced optimization techniques, we aim to enhance decision-making in supplier selection.

#### 3.1 Problem description

Despite the progress made in sustainable supplier selection and order allocation models, particularly those utilizing MCDM and hybrid approaches, existing frameworks often fail to fully integrate probabilistic risk considerations, especially in the context of disruption-prone environments. Many current methods rank or prioritize suppliers based on sustainability metrics but lack mechanisms for dynamic allocation under uncertainty. This gap is particularly critical in high-risk industries, such as automotive manufacturing, where supplier performance is vulnerable to disruptions. To address this issue, we propose a chance-constrained GP model that incorporates sustainability criteria, cost factors, and disruption risk using Value-at-Risk

(VaR)-based quantification. Our framework enables decision-makers to balance conflicting goals while accounting for uncertainty in supplier performance.

Choosing a supplier is a multi-criteria decision-making process that involves both qualitative and quantitative elements. We propose a chance-constrained GP method for choosing the best sustainable provider under disruption risk.

The proposed model integrates three key objectives:

- (1) *Cost Minimization*: Includes both variable and fixed costs, with total procurement cost modeled under uncertainty using a chance-constrained Value-at-Risk (VaR) formulation.
- (2) *Sustainability Maximization*: Encourages sourcing from suppliers with high sustainability scores, modeled via a probabilistic sustainability threshold constraint.
- (3) *Risk Mitigation*: Achieved by limiting the probability of cost overrun and ensuring supplier diversification.

The integration of VaR allows the model to quantify the maximum potential procurement loss at a given confidence level  $(1-\alpha)$ , capturing supply-side financial risk. The chance-constrained nature of the model ensures that procurement decisions respect probabilistic thresholds, enabling robust decision-making under uncertainty. Table 2 summarizes the symbols, descriptions, and units or types of all decision variables, parameters, and indices employed in the mathematical formulation.

The three goals of the suggested optimization model are to minimize the total cost of purchases, source as much as feasible from suppliers who have demonstrated exceptional sustainability, and lower the supply risk. Refers to Kellner and Utz (2019), the total cost of purchase, which consists of both variable and fixed expenses, is reduced by using Equation (1).

$$\min \sum_{i=1}^N (cv_i * x_i * D + cf_i * y_i) \quad (1)$$

The computation of variable costs is based on the selling prices per unit ( $cv_i$ ). Costs associated with establishing and preserving the relationship with the provider for the duration of the study are known as fixed costs, or ( $cf_i$ ).

In this paper, we developed a novel equation by combining the chance-constrained technique, confidence levels  $\alpha$ , and the concept of Value at Risk (VaR), taking inspiration from the formulation presented by Equation (1). With this revised formulation, procurement cost unpredictability is better managed. With a confidence level of  $(1-\alpha)$ , the probabilistic constraint included in the revised equation guarantees that the overall costs will not above a predetermined threshold (VaR). This strategy guarantees the efficient and safe management of supplier relationships while enabling greater prediction of the financial risks related to cost variations.

$$P\left(\sum_{i=1}^N (cf_i * x_i * D + cv_i * y_i) \leq VaR_s\right) \geq (1 - \alpha) \quad (2)$$

Assuming that returns have a normal distribution, the Value at Risk (VaR) can be computed parametrically using the suppliers' variance-covariance matrix. The linkages between the risks of various suppliers are captured by the variance-covariance matrix.

The overall portfolio variance can be determined by combining the elements of this matrix with the weights allocated to each supplier. The volatility of the portfolio (the square root of the variance) is then multiplied by the total value of the portfolio plus a predetermined confidence factor to determine the VaR.

**Table 2.** Integrated overview of criteria, parameters, and notations applied in the model

| Symbol                            | Description                                                                  | Type/Unit                  |
|-----------------------------------|------------------------------------------------------------------------------|----------------------------|
| $i = 1, ..., N$                   | Index and set of suppliers                                                   | Integer/Set                |
| $j \in J$                         | Index and set of demand items (or locations/components, based on your model) | Integer/Set                |
| $d_j$                             | Demand for item/component $j$                                                | Units                      |
| $c_i$                             | Unit cost offered by the supplier $i$                                        | Monetary units             |
| $s_i$                             | Sustainability score of the supplier $i$                                     | Dimensionless              |
| $r_i$                             | Disruption risk level associated with the supplier $i$                       | Probability/Risk score     |
| $x_i$                             | Quantity (or proportion) of demand allocated to supplier $i$                 | Real number [0,1] or units |
| $y_i$                             | Binary decision variable: 1 if supplier $i$ is selected; 0 otherwise         | Binary {0, 1}              |
| $B$                               | Total procurement budget                                                     | Monetary units             |
| $E$                               | Maximum allowed environmental impact                                         | Environmental units        |
| $\alpha$                          | Risk tolerance level (confidence level for VaR constraint)                   | Real number [0,1]          |
| $VaR$                             | Value-at-Risk threshold                                                      | Probability-weighted loss  |
| $\lambda_1, \lambda_2, \lambda_3$ | Weighting coefficients for goals (cost, sustainability, risk)                | Real numbers               |
| $Z$                               | Objective function value                                                     | Composite/dimensionless    |

**Note(s):**  $W_1$ : The weight for the sustainability goal  
 $W_2$ : The weight for the risk goal

**Decision variables**  
 $x_i \in [0, 1]$ : A fraction of the whole quantity ordered from the provider  $i$   
 $y_i \in [0, 1]$ : Takes 1 if supplier  $i$  is included in the portfolio, 0 otherwise

**Parameters**  
 $Cv_i \in R$  The selling price of supplier  $i$  (variable cost) per unit  
 $Cf_i \in R$  Fixed cost associated with sourcing from a supplier  $i$   
 $log_i \in [0, 1]$ : A measure of the logistical performance of supplier  $i$   
 $srs_i \in [0, 1]$ : Sustainability performance of Supplier  $i$  (sustainability rating score)  
 $CAP_i \in [0, 1]$ : The portion of the whole demand that can be sourced from the supplier's capacity  $i$   
 $MOQ_i \in [0, 1]$ : Minimum order quantity allocated the supplier  $i$  (as a % of the total demand)  
 $strat_i \in \{0, 1\}$ : Binary variable, 1 if supplier  $i$  is a "strategic" supplier, 0 otherwise  
 $reg_i \in \{0, 1\}$ : Binary variable, 1 if supplier  $i$  is a regional supplier, 0 otherwise  
 $D \in R$ : Total Demand  
 $B \in R$ : Allocated Budget  
 $L \in [0, 1]$ : Minimum logistics service desired  
 $N_{max} \in Z$ : The maximum of selected suppliers in the portfolio  
 $N_{str} \in Z$ : The number of strategic suppliers in the portfolio  
 $N_{reg} \in Z$ : The number of regional suppliers in the portfolio

**Source(s):** Authors' elaboration

Utilizing a variance-covariance matrix, supplier sustainability scores, and a predetermined confidence level, I computed the Value-at-Risk (VaR) using R software. The greatest possible loss was calculated by calculating the return and volatility of the portfolio.

Drawing inspiration from [Kelner and Utz's \(2019\)](#) equation ([Equation 3](#)) that aims to maximize the proportion of orders placed with suppliers who demonstrate high sustainability performance:

$$Max \left( \sum_{i=1}^N srs_i * x_i \right) \tag{3}$$

We have expanded this formulation to include chance-constrained programming and confidence levels  $\alpha$ . With a confidence level of  $(1 - \alpha)$ , the equation's revised formulation

guarantees that the total of sustainability ratings. The chosen suppliers'  $srs_i$  will either reach or surpass a predetermined sustainability level  $SR_s$ . By taking into consideration supplier performance risks, this probabilistic method ensures that suppliers with high sustainability ratings are given priority and offers a more resilient framework.

$$P\left(\sum_{i=1}^N srs_i * x_i \geq SR_s\right) \geq (1 - \alpha) \quad (4)$$

Following [Messaoudi et al. \(2017\)](#), these chance constraints can be rewritten as follows:

$$\max \sum_{i=1}^N srs_i * x_i \geq E(SR_s) + \varphi^{-1}(1 - \alpha) * \sigma(SR_s) \quad (5)$$

$$\sum_{i=1}^N (cf_i * x_i * D + cv_i * y_i) \leq E(VaR_s) + \varphi^{-1}(1 - \alpha) * \sigma(VaR_s) \quad (6)$$

### 3.2 Model assumptions and justifications

To ensure both transparency and methodological rigor, several assumptions underlie the proposed model. First, supplier cost fluctuations and performance variations are considered to follow a normal distribution. This choice is common in financial risk modeling and facilitates the use of parametric Value-at-Risk (VaR) computations. Second, the risks linked to different suppliers are treated as statistically independent, which simplifies the construction of the variance–covariance matrix and aligns with the logic of multi-sourcing strategies in supply chain risk analysis. Third, supplier-specific parameters such as capacities, sustainability scores, and cost structures are assumed to remain fixed throughout the planning horizon, as reflected in the data obtained from the case study. Finally, the model accounts for varying degrees of risk aversion by incorporating multiple confidence levels ( $\alpha$ ), ranging from 0.25 to 0.99, thereby enabling a comprehensive sensitivity analysis across different tolerance thresholds.

These assumptions strike a balance between analytical tractability and practical applicability, and are in line with methodologies used in prior studies such as [Kellner and Utz \(2019\)](#), and [Moheb-Alizadeh and Handfield \(2019\)](#).

### 3.3 Model formulation

This study considers a centralized, single-tier procurement setting in which a focal firm (the buyer) must choose and manage a portfolio of suppliers to meet product demand. The supply base comprises multiple suppliers with varying cost structures, environmental performance scores, and degrees of exposure to disruption risk. Within the proposed framework, the buyer must address three interrelated decisions. The first concerns supplier selection, which requires evaluating potential partners in terms of cost efficiency, sustainability performance, and vulnerability to disruption risk. The second involves order allocation, namely determining how the overall demand should be distributed among the chosen suppliers while ensuring compliance with budgetary limits and environmental thresholds. The third decision relates to risk management, where the buyer establishes acceptable levels of exposure to supply disruptions by applying a Value-at-Risk (VaR) constraint.

The model assumes that supplier characteristics such as prices, sustainability ratings, capacities, and historical reliability are known and fixed during the planning horizon. External disruptions (e.g. political instability, natural hazards) are modeled probabilistically. The objective is to support the buyer in designing a procurement strategy that balances cost-effectiveness, sustainable sourcing, and resilience against uncertainty.

## MSCRA

The chance-constrained formulation follows the structure introduced in [Rostami et al. \(2023\)](#) and [Jia et al. \(2020\)](#). Equation (2) adapts the weighted goal programming approach discussed in [Ben Abdallah et al. \(2024\)](#).

Our model presents a novel paradigm for supplier selection that incorporates risk reduction, sustainability, and cost-effectiveness. The following is a presentation of our suggested chance-constrained multiple aims programming:

$$\text{Min } Z = W_1^- \delta_1^- + W_2^+ \delta_2^+ \quad (7)$$

Subject to

$$\sum_{i=1}^N srs_i * x_i - E(SR_s) - \varphi^{-1}(1 - \alpha) * \sigma(SR_s) + \delta_1^- - \delta_1^+ = 0 \quad (8)$$

$$\sum_{i=1}^N (cf_i * x_i * D + cv_i * y_i) - E(VaR_s) - \varphi^{-1}(1 - \alpha) * \sigma(VaR_s) - \delta_2^+ + \delta_2^- = 0 \quad (9)$$

$$\text{Fulfillment of demand } \sum_i x_i = 1 \quad (10)$$

$$\text{Capacity of suppliers } x_i \leq CAP_i * y_i \forall i \quad (11)$$

$$\text{Minimum quantity ordered } x_i \geq MOQ_i * y_i \forall i \quad (12)$$

$$\text{Budget allotted } \sum_{i=1}^N (cf_i * x_i * D + cv_i * y_i) \leq B \quad (13)$$

$$\text{Logistics backup service. } L \leq \sum_i Log_i * x_i \quad (14)$$

$$\text{Dual sourcing } 2 \leq \sum_i y_i \quad (15)$$

$$\text{The maximum number of providers } \sum_i y_i \leq N_{max} \quad (16)$$

$$\text{The number of strategic suppliers } \sum_i y_i * \text{strat}_i \geq N_{str} \quad (17)$$

$$\text{The number of regional suppliers } \sum_i y_i * \text{reg}_i \geq N_{reg} \quad (18)$$

Equations (7)–(18) define a multi-objective optimization model for supplier selection, incorporating sustainability, cost efficiency, operational feasibility, and risk mitigation. Equation (7) defines the objective function which seeks to minimize total cost while maximizing sustainability and minimizing risk. Equation (8) seeks to optimize the allocation of orders to suppliers with superior sustainability performance. This assessment is quantified using the Sustainability Rating Score (SRS), which ranges from 0 to 1, where higher values indicate stronger adherence to sustainability principles. Equation (9) minimizes total

procurement expenditures, encompassing both variable and fixed costs. The unit price  $cv_i$  directly influences variable costs, while fixed costs ( $cf_i$ ) account for the expenses incurred in establishing and maintaining supplier relationships throughout the evaluation period. Equation (10) ensures that the total demand of the purchasing organization is fully satisfied. The aggregate supply from all selected suppliers must match the firm's total procurement requirements. Equation (11) imposes an upper limit on the order volume allocated to each supplier, ensuring that no supplier receives orders exceeding its maximum supply capacity. Equation (12) enforces compliance with minimum order quantity ( $MOQ$ ) constraints, ensuring that no supplier is assigned an order below its predefined  $MOQ$  threshold. Equation (13) imposes a financial constraint, ensuring that the total procurement expenditure does not exceed the organization's budgetary limits. Equation (14) maintains a minimum required logistics service level across the selected supplier portfolio, ensuring that overall logistics performance meets or exceeds a predefined threshold. Equation (15) enforces supplier diversification by mandating the inclusion of at least two suppliers in the portfolio. This requirement enhances supply chain resilience by mitigating risks associated with supplier failure and fostering competitive dynamics that may lead to more favorable contractual terms. Equation (16) sets an upper bound on the total number of selected suppliers, balancing supply chain complexity and management costs while ensuring procurement efficiency. Equation (17) guarantees the inclusion of at least  $N_{str}$  strategic suppliers within the portfolio. These suppliers possess critical competencies such as expertise in research and development (R&D), adaptability to fluctuations in demand, or specialized manufacturing capabilities. This constraint can be further decomposed to address specific strategic attributes, ensuring the incorporation of suppliers with essential domain expertise.

Equation (18) mandates the inclusion of at least  $N_{reg}$  regional suppliers to enhance supply chain responsiveness. Regional suppliers contribute to reducing lead times and improving adaptability to localized market fluctuations. Moreover, their presence enhances supply chain resilience by introducing geographic diversification, mitigating risks associated with regional disruptions. It ensures that a *minimum number of suppliers* from a specific geographic region are selected, in line with supply chain diversification or localization policies. This promotes resilience and supports sustainability initiatives in underrepresented regions.

The portfolio-level Value-at-Risk (VaR) for procurement cost is computed using:  $VaR = Z_{1-\alpha} \cdot \sqrt{W^T * \sum w * P}$

Where:

- (1)  $Z_{1-\alpha}$  is the quantile of the standard normal distribution corresponding to the confidence level.
- (2)  $\sum$  is the variance-covariance matrix of supplier cost returns.
- (3)  $W$  is the vector of allocation weights to each supplier.
- (4)  $P$  is the total portfolio value.

This formulation ensures that the cost-related risk is bounded by a pre-defined threshold under a specified confidence level, supporting risk-aware procurement planning.

#### 4. Application: a real-life instance study

The case study data is grounded in the empirical framework presented by Kellner and Utz (2019), which examined sustainable supplier management in the automotive industry. While the core structure and parameter ranges follow the original study, some values were anonymized or simulated to protect commercial confidentiality. We use an example from the real world of the automotive industry to show how the suggested approach can be applied. The case focuses on a leading luxury car OEM in Germany that faces the difficulty of choosing suppliers and allocating orders for embedded navigation systems.

The following characteristics apply to the general purchase scenario in addition to the supplier-specific information: There is a demand for 100,000 units, and the maximum budget that can be given is  $D$ . The largest budget that can be allocated ( $B$ ) is \$14,000,000, while the minimum logistics service needed ( $L$ ) is 90%. Furthermore, the minimum number of strategic ( $N_{str}$ ) and regional ( $N_{reg}$ ) suppliers is 1, while the maximum number of suppliers ( $N_{max}$ ) is 4.

The sustainability rating scores of the eight vendors are shown in Table 3. This study sets itself apart by using sustainability ratings obtained from a self-assessment questionnaire in line with the Global Automotive Sustainability Guiding Principles (GASGP), rather than from self-created performance indicators. The benchmark for assessing supplier sustainability in the automotive sector is the questionnaire, which was created in 2014. This commonly used self-assessment method serves as the foundation for the sustainability performance ratings of the eight suppliers that are the subject of the inquiry.

Compared to more sophisticated techniques covered in the literature, like TOPSIS, PROMETHEE, DEA, AHP, or ANP, self-assessment may appear less sophisticated, yet it has several important benefits. These include a uniform methodology among suppliers, simplicity of data gathering, and conformity to industry-wide sustainability guidelines, all of which improve the procurement process's comparability and openness.

#### 4.1 Results

The optimization results for different weight configurations of  $W_1^-$  and  $W_2^+$  across various values of  $\alpha$  are presented in Tables 4–6. These results provide insights into how the objective function, the number of selected suppliers, and the values of  $\delta_1^-$  and  $\delta_2^+$  vary with different risk and sustainability weight allocations. The analysis allows us to assess the trade-offs between minimizing costs, prioritizing sustainable suppliers, and mitigating supply risks under uncertainty.

Figure 1 illustrates the variation of  $\delta_1^-$  (red line) and  $\delta_2^+$  (green line) across different  $\alpha$  values for  $W_1^- = 0.4$  and  $W_2^+ = 0.6$ . The green line remains relatively stable, indicating minimal fluctuation in  $\delta_2^+$ , while the red line shows an increasing trend initially, reaching a peak, and then slightly decreasing. This suggests that  $\delta_1^-$  is more sensitive to changes in  $\alpha$ , reflecting the impact of risk and sustainability trade-offs in the optimization model.

$\delta_1^-$  Changes based on the significance level ( $\alpha$ ), increasing up to  $\alpha = 0.75$  and then decreasing afterward. The relationship between  $\alpha$  and  $U1$  is not straightforward, with  $\delta_1^-$  rising up to a certain point ( $\alpha = 0.75$ ) and then tapering off as the  $\alpha$  level increases. This kind of relationship is indicative of a situation where moderate confidence provides a more substantial result or effect, while higher confidence leads to diminishing returns or a more cautious estimate.

$\delta_2^+$  A fixed value (1,017,000) that is independent of  $\alpha$ , representing a baseline that does not change with confidence level adjustments.

*Fct objective:* A constant objective (610,200) that doesn't change with  $\alpha$ , which might indicate a set goal that remains the same despite adjustments in  $\alpha$ .

Overall, the optimization results confirm the ability of the proposed framework to balance procurement cost, sustainability performance, and exposure to disruption risk within supplier portfolio design. The solutions consistently favored a mix of regionally diverse and environmentally responsible suppliers, demonstrating that diversification and sustainability thresholds can enhance robustness against uncertainty. Importantly, the incorporation of Value-at-Risk (VaR) constraints allowed disruption risk to be contained without producing significant cost increases. These outcomes highlight that multi-criteria optimization combined with probabilistic risk modeling can deliver procurement strategies that are both resilient and sustainable. For practitioners, this implies that firms, particularly in high-risk sectors such as automotive and electronics, can strengthen resilience and regulatory compliance by embedding risk-sensitive, sustainability-oriented models into supplier selection processes.

**Table 3.** An overview of the attributes of all suppliers

|                                        | S-1       | S-2       | S-3       | S-4       | S-5       | S-6       | S-7        | S-8       |
|----------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|-----------|
| Variable costs ( <i>cv</i> )           | 115       | 100       | 108       | 107       | 113       | 110       | 109        | 101       |
| Fixed costs ( <i>cf</i> )              | 2.00 E+06 | 9.00 E+06 | 1.00 E+06 | 1.10 E+06 | 1.50 E+06 | 1.10 E+06 | 1.150 E+06 | 9.50 E+06 |
| Logistics service ( <i>log</i> )       | 0.992     | 0.753     | 0.859     | 0.878     | 0.979     | 0.906     | 0.858      | 0.799     |
| Sustainability score ( <i>srs</i> )    | 0.82      | 0.32      | 0.72      | 0.48      | 0.53      | 0.69      | 0.79       | 0.41      |
| Min order qty ( <i>MOQ</i> )           | 0.1       | 0.1       | 0.1       | 0.1       | 0.1       | 0.1       | 0.1        | 0.1       |
| Capacity ( <i>CAP</i> )                | 1.0       | 0.7       | 1.0       | 1.0       | 1.0       | 0.6       | 1.0        | 1.0       |
| Strategic sup ( <i>strat</i> )         | 1         | 0         | 0         | 1         | 1         | 1         | 1          | 0         |
| Regional sup ( <i>reg</i> )            | 1         | 0         | 1         | 0         | 1         | 1         | 0          | 0         |
| <b>Source(s):</b> Authors' elaboration |           |           |           |           |           |           |            |           |

**Table 4.** Optimization results for  $W_1^- = 0.4$  and  $W_2^+ = 0.6$  across different values of  $\alpha$

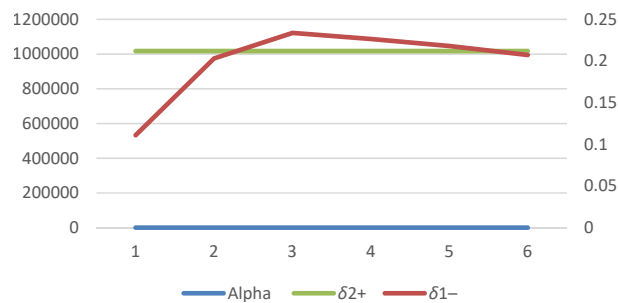
| $\alpha$                        | 0.25      | 0.5       | 0.75      | 0.9       | 0.95      | 0.99      |
|---------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Phi-1                           | -0.6745   | 0         | 0.674     | 1.28      | 1.645     | 2.33      |
| Fct objective                   | 610,200   | 610,200   | 610,200   | 610,200   | 610,200   | 610,200   |
| Number of selected suppliers    | 4         | 4         | 4         | 4         | 4         | 4         |
| $\delta_1^-$                    | 0.1109146 | 0.2030000 | 0.2336724 | 0.2263001 | 0.2179721 | 0.2072413 |
| $\delta_2^+$                    | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 |
| Source(s): Authors' elaboration |           |           |           |           |           |           |

**Table 5.** Optimization results for  $W_1^- = 0.6$  and  $W_2^+ = 0.4$  across different values of  $\alpha$

| $\alpha$                        | 0.25      | 0.5      | 0.75      | 0.9       | 0.95      | 0.99      |
|---------------------------------|-----------|----------|-----------|-----------|-----------|-----------|
| Phi-1                           | -0.6745   | 0        | 0.674     | 1.28      | 1.645     | 2.33      |
| Fct objective                   | 406800.1  | 406800.1 | 406800.1  | 406800.1  | 406800.1  | 406800.1  |
| Number of selected supplier     | 4         | 4        | 4         | 4         | 4         | 4         |
| $\delta_1^-$                    | 0.1109146 | 0.203    | 0.2336724 | 0.2263001 | 0.2179721 | 0.2072413 |
| $\delta_2^+$                    | 1,017     | 1,017    | 1,017     | 1,017     | 1,017     | 1,017     |
| Source(s): Authors' elaboration |           |          |           |           |           |           |

**Table 6.** Optimization results for  $W_1^- = 0.5$  and  $W_2^+ = 0.5$  across different values of  $\alpha$

| $\alpha$                        | 0.25      | 0.5       | 0.75      | 0.9       | 0.95      | 0.99      |
|---------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Phi-1                           | -0.6745   | 0         | 0.674     | 1.28      | 1.645     | 2.33      |
| Fct objective                   | 508500.1  | 508500.1  | 508500.1  | 508500.1  | 508500.1  | 508500.1  |
| Number of selected supplier     | 4         | 4         | 4         | 4         | 4         | 4         |
| $\delta_1^-$                    | 0.1109146 | 0.2030000 | 0.2336724 | 0.2263001 | 0.2179721 | 0.2072413 |
| $\delta_2^+$                    | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 | 1,017,000 |
| Source(s): Authors' elaboration |           |           |           |           |           |           |



**Figure 1.** Variation of  $\delta_1^-$  and  $\delta_2^+$  for  $W_1^- = 0.4$  and  $W_2^+ = 0.6$  across different values of  $\alpha$ . Source: Authors' elaboration

#### 4.2 Result analysis

$\alpha$ : Used in statistical risk assessment and decision-making, it symbolizes various confidence interval levels. A 75% confidence level, for instance, is represented by  $\alpha = 0.25$ , and a 95% confidence level by  $\alpha = 0.95$ .

Z-scores corresponding to the various  $\alpha$  values are represented by Phi-1. Z-scores indicate the number of standard deviations that a given value deviates from the mean. A 95% confidence level, for instance, is associated with a Z-score of 1.645. The Z-scores (Phi-1) rise and move away from 0 as  $\alpha$  rises, indicating an increase in confidence. Higher confidence intervals typically include more data points; thus, this is to be expected. A higher Z-score is indicative of a lower risk tolerance (i.e. more conservative actions), and the Phi-1 values can be used to measure risk tolerance.

Based on all tables (Tables 4-6), for every  $\alpha$  level, the value of the goal function remains constant at 610,200 (Table 4). This could mean that, for this particular situation, changes in the confidence level have no effect on the optimization model. Regardless matter the level of risk accepted, the same goal is accomplished. The goal function's stability implies that, depending on what the objective function represents, different levels of confidence have no effect on the cost, profit, or performance indicator. This could indicate that there is minimal diversity in outcomes between the designated suppliers (four in all cases) or that the system is resilient to varying degrees of risk.

Across all  $\alpha$  levels, the number of selected suppliers remains constant at 4. Regardless of the risk threshold ( $\alpha$ ), the model has determined that 4 providers are the best option for minimizing objective function, as evidenced by the fact that the number of selected suppliers remains constant. This consistency may indicate that other configurations are less effective and that these four providers are essential to fulfilling the needs of the model.

The rise in  $\delta_1^-$  values with increasing  $\alpha$  (from 0.1109 at  $\alpha = 0.25$  to 0.2072 at  $\alpha = 0.99$ ) suggests that risk perception has a minor effect on resource allocation or supplier utilization. A relationship between increased confidence levels and the performance or utilization of specific resources or suppliers is suggested by the minor increase in  $\delta_1^-$  values as  $\alpha$  rises. This slow increase could be a sign of increased resource use or a change in allocation tactics as risk tolerance evolves.

Across all  $\alpha$  values,  $\delta_1^-$  stays at 1,017,000. This implies that, regardless of the degree of confidence in the risk estimations, some variables or resources) remain constant.

#### 4.3 Discussion of methodological implications

The consistent outcomes across different  $\alpha$  values confirm the robustness of the model under varying risk scenarios. This suggests that the selected suppliers possess structural strengths such as high sustainability scores and consistent logistics service that render them optimal choices across a range of risk tolerance levels. The use of probabilistic constraints and VaR enables firms to quantify and control financial risk exposure in supplier selection.

**4.3.1 Managerial and theoretical implications. Managerial Implications:** The findings of this study provide actionable insights for procurement and supply chain professionals navigating uncertainty and sustainability trade-offs. One of the key takeaways is the model's ability to yield stable and consistent supplier recommendations across a range of risk tolerance levels. This robustness allows decision-makers to customize their sourcing strategies based on organizational priorities such as minimizing cost volatility or complying with environmental targets without compromising decision quality.

By adjusting the confidence level parameter ( $\alpha$ ), managers can tailor the degree of risk aversion embedded in the solution. This flexibility is particularly relevant in industries like automotive or electronics, where supply disruptions can have cascading operational effects. The inclusion of Value-at-Risk (VaR) constraints in the model equips firms with a practical tool to assess and cap potential losses from supplier failures, in alignment with internal risk governance or regulatory requirements.

Furthermore, the framework integrates measurable sustainability criteria such as emissions levels or supplier certifications into the decision process. This enhances its alignment with real-world procurement practices where sustainability audits and ESG performance are increasingly part of supplier evaluation protocols. Overall, the proposed model serves as a decision support system that can strengthen supplier portfolio design, support compliance objectives, and improve resilience in global sourcing environments.

*Theoretical Implications:* From a theoretical perspective, this work contributes to the advancement of supplier selection methodologies by embedding stochastic reasoning within a multi-objective optimization framework. Unlike conventional approaches that treat cost, risk, or sustainability in isolation, the proposed model addresses the interplay between these dimensions under uncertainty.

The integration of chance-constrained programming with a Value-at-Risk structure provides a novel linkage between operations research and financial risk assessment. This hybrid formulation moves beyond deterministic or single-scenario models and responds to recent scholarly calls for richer, uncertainty-aware decision tools in procurement science. In doing so, the study extends prior research (e.g. [Moheb-Alizadeh and Handfield, 2018](#); [Kellner and Utz, 2019](#)) by offering a quantitative structure that reflects the complexity and interdependence of real-world sourcing decisions.

The methodological contribution lies not only in the formulation of the model, but in its practical interpretability and scalability. It can be adapted to various industrial settings, making it a versatile tool for academic researchers and practitioners seeking to balance sustainability goals with cost efficiency and operational reliability.

## 5. Conclusion

This research presents an integrated decision-support framework for sustainable supplier selection and order allocation under uncertainty, leveraging a chance-constrained multi-objective optimization model. By embedding Value-at-Risk (VaR) and probabilistic constraints into the formulation, the approach moves beyond traditional deterministic models and offers a risk-sensitive, sustainability-aligned procurement solution.

Applied to a case study in the automotive industry, the model demonstrated strong practical relevance. The consistency of supplier choices across a range of confidence levels highlights the robustness of the solution space, while controlled parameter sensitivity indicates the model's resilience to variations in uncertainty levels. This assures procurement professionals that stable sourcing strategies can be maintained, even in the face of external disruptions or regulatory shifts.

From a managerial perspective, the proposed model acts as a structured tool to navigate complex trade-offs between cost minimization, supply chain risk mitigation, and environmental responsibility. Its compatibility with standard sustainability metrics and its consideration of budgetary and capacity constraints ensure operational feasibility across a range of industrial settings. The ability to incorporate firm-specific risk tolerance also empowers decision-makers to tailor sourcing strategies based on strategic priorities and market conditions.

From a theoretical standpoint, the study contributes to the advancement of supply chain modeling by integrating concepts from financial risk management into sustainable procurement optimization. The framework extends classical goal programming by incorporating probabilistic constraints and VaR measures, offering a novel mathematical structure that better captures real-world decision environments. This interdisciplinary linkage enhances the methodological landscape of supplier selection under uncertainty.

While the model offers considerable flexibility and relevance, some assumptions, such as the normality of disruption risks and reliance on pre-assessed supplier scores, may constrain generalizability. Future research should aim to relax these assumptions by integrating real-time data streams, adaptive learning mechanisms, and dynamic supplier evaluations.

Moreover, expanding the model to multi-tier supply networks and incorporating circular economy dimensions could further strengthen its contribution to sustainable development goals. Future enhancements to the model could integrate bio-inspired metaheuristics such as ant colony or genetic algorithms to address scalability and computational complexity (e.g. [Euchi et al., 2016](#)).

In conclusion, this study delivers a scientifically grounded and practically applicable approach to supplier selection and order allocation in uncertain and sustainability-driven supply chains. It equips organizations with a robust decision-making tool that balances economic performance, environmental goals, and resilience, key imperatives in today's rapidly evolving global supply networks.

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**Corresponding author**

Jalel Euchí can be contacted at: [jalel.euchi@fsegs.rnu.tn](mailto:jalel.euchi@fsegs.rnu.tn)